

Multi-modality Learning

Jian Tang

HEC Montreal

Mila-Quebec AI Institute

CIFAR AI Research Chair

Email: jian.tang@hec.ca



Outline

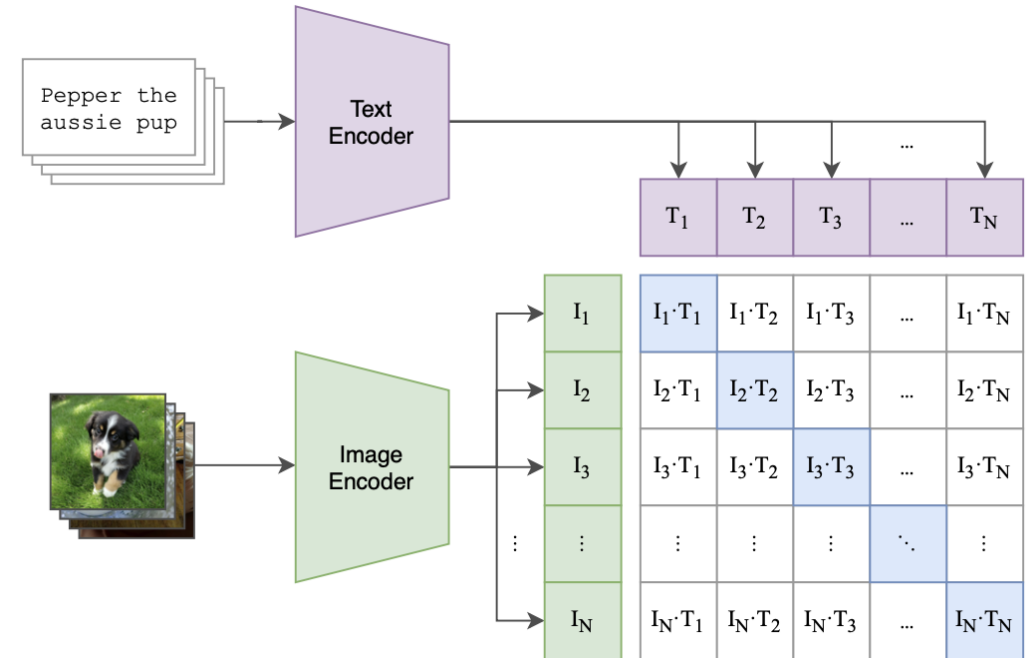
- CLIP: Joint Text and Image Embedding
- DALL-E-2: Generating Images from Text
- LLaVA: Multi-modality Conversation

CLIP: Joint image and text representation learning (Radford et al. 2021)

- Goal: learning SOTA image representations
 - Expensive to obtain a large amount of labeled images
 - Learning image representations from supervision contained in natural language
- Training data: 400 million (image, text) pairs collected from Internet

CLIP (Radford et al. 2021)

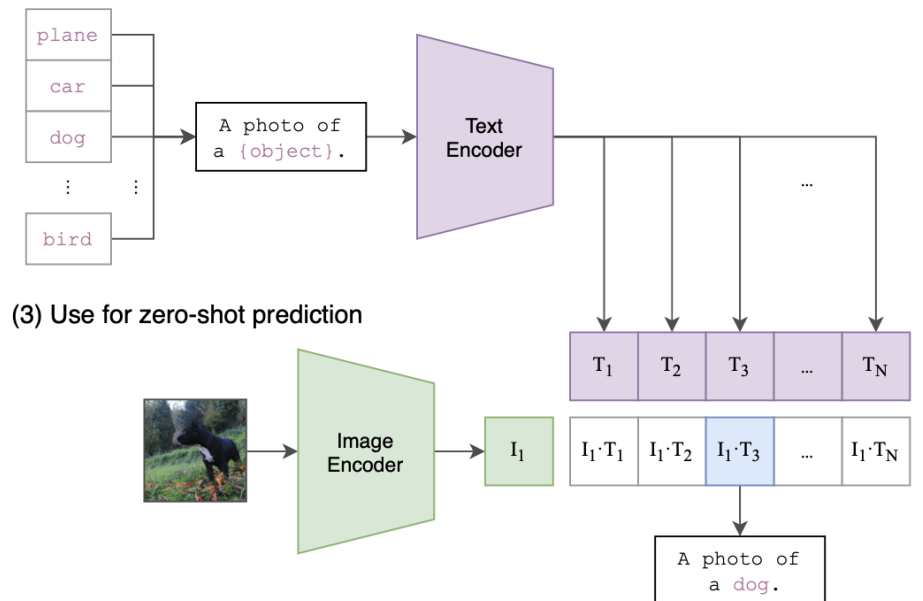
- Contrastive Pre-training
 - A text encoder based on a Transformer language mod
 - An image encoder
- Jointly train an image encoder and a text encoder to predict the correct pairings of a batch of (image, text) training examples



Zero-shot Image Classification

- Use the names of all the classes in the dataset as the set of potential text pairings
- Predict the most probable (image, text) pair according to CLIP
 - Compute the class embedding with the text encoder
 - Compute the image embedding with the image encoder

(2) Create dataset classifier from label text



$$p(y_i|x) = \frac{\exp(\cosine(vec(x), vec(y_i)))}{\sum_{j=1}^K \exp(\cosine(vec(x), vec(y_j)))}$$

Zero-shot Performance

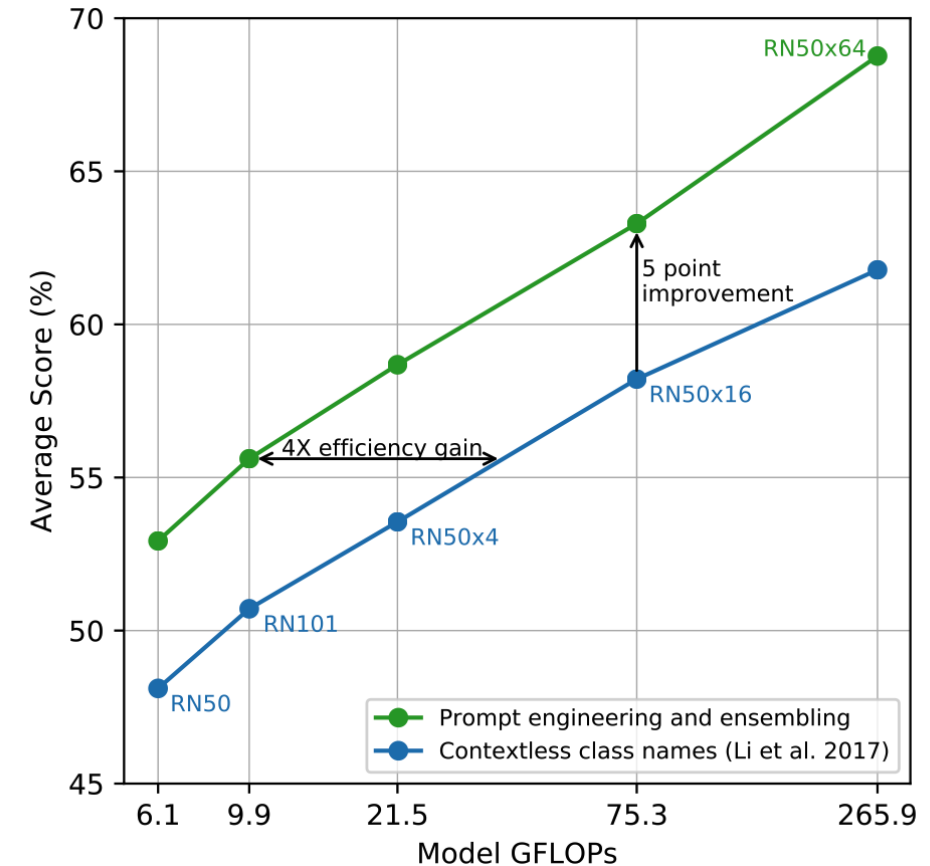
- Significant performance improvement compared to previous methods

	aYahoo	ImageNet	SUN
Visual N-Grams	72.4	11.5	23.0
CLIP	98.4	76.2	58.5

Table 1. Comparing CLIP to prior zero-shot transfer image classification results. CLIP improves performance on all three datasets by a large amount. This improvement reflects many differences in the 4 years since the development of Visual N-Grams ([Li et al., 2017](#)).

Prompt Engineering and Ensembling

- Class encoder with different prompts:
 - Only with the class names
 - With the prompt “A photo of a {label}.”
- Ensemble multiple classifiers with different context prompts, e.g.,
 - “A photo of a big {label}”
 - “A photo of a small {label}”
- The ensemble is constructed in the embedding space



Zero-shot CLIP is competitive with a fully supervised baseline

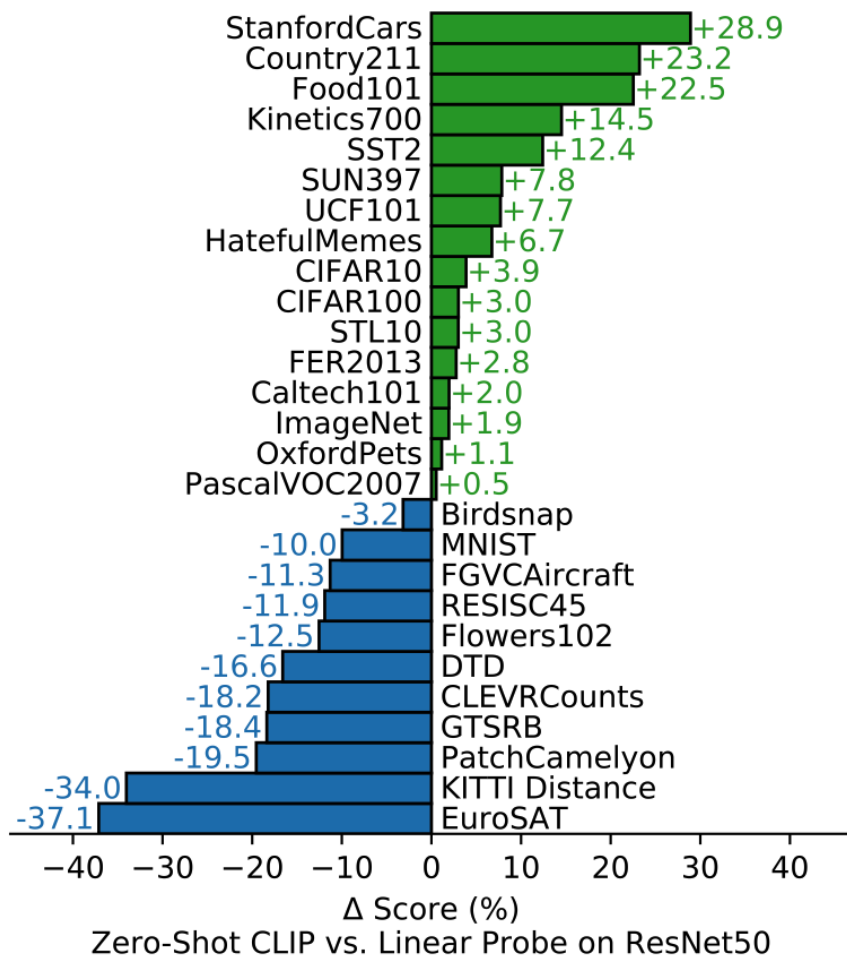


Figure 5. Zero-shot CLIP is competitive with a fully supervised baseline. Across a 27 dataset eval suite, a zero-shot CLIP classifier outperforms a fully supervised linear classifier fitted on ResNet-50 features on 16 datasets, including ImageNet.

Zero-shot CLIP outperforms few-shot linear probes

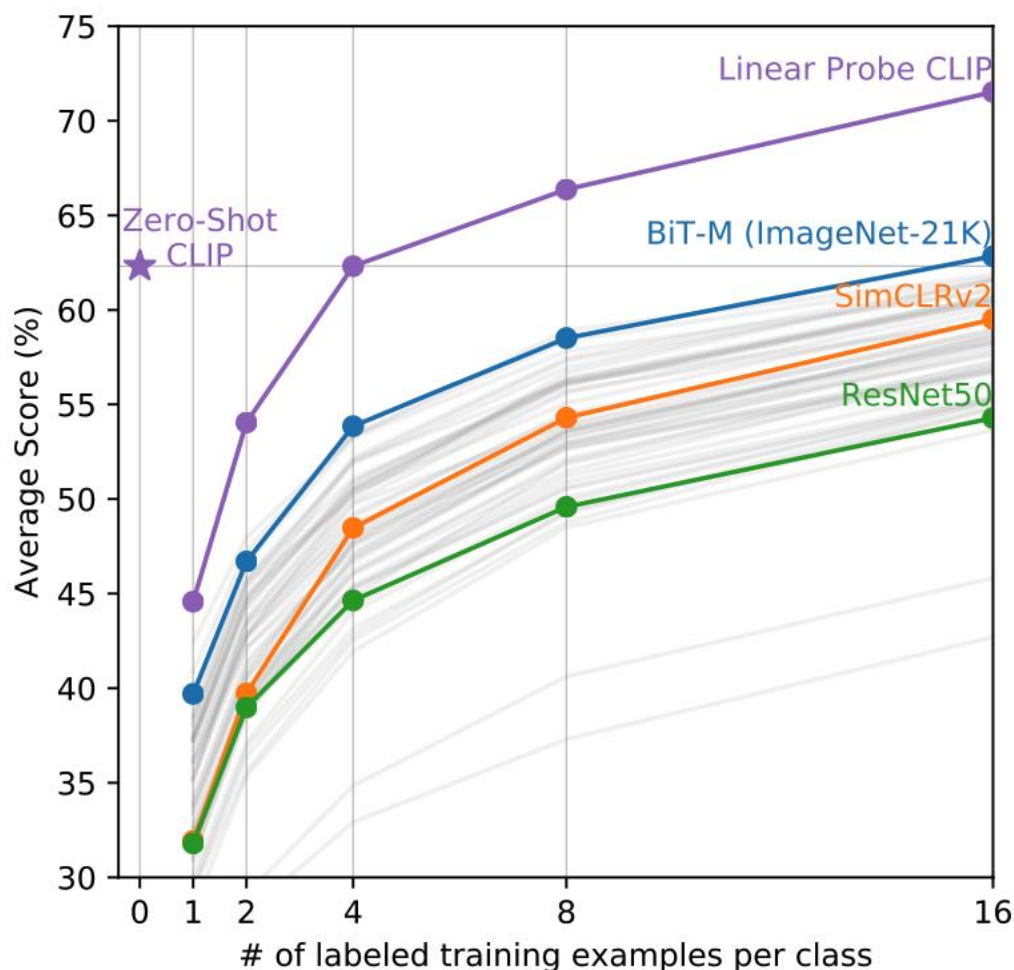


Figure 6. Zero-shot CLIP outperforms few-shot linear probes. Zero-shot CLIP matches the average performance of a 4-shot linear classifier trained on the same feature space and nearly matches the best results of a 16-shot linear classifier across publicly available models. For both BiT-M and SimCLRv2, the best performing model is highlighted. Light gray lines are other models in the eval suite. The 20 datasets with at least 16 examples per class were used in this analysis.

Implementation

```
# image_encoder - ResNet or Vision Transformer
# text_encoder  - CBOW or Text Transformer
# I[n, h, w, c] - minibatch of aligned images
# T[n, l]       - minibatch of aligned texts
# W_i[d_i, d_e] - learned proj of image to embed
# W_t[d_t, d_e] - learned proj of text to embed
# t            - learned temperature parameter

# extract feature representations of each modality
I_f = image_encoder(I) #[n, d_i]
T_f = text_encoder(T)  #[n, d_t]

# joint multimodal embedding [n, d_e]
I_e = l2_normalize(np.dot(I_f, W_i), axis=1)
T_e = l2_normalize(np.dot(T_f, W_t), axis=1)

# scaled pairwise cosine similarities [n, n]
logits = np.dot(I_e, T_e.T) * np.exp(t)

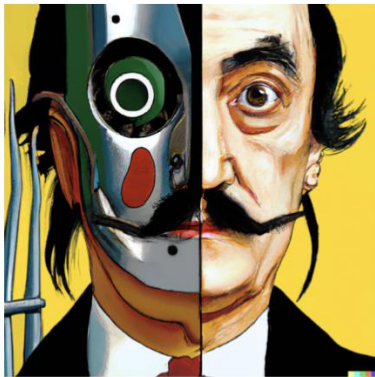
# symmetric loss function
labels = np.arange(n)
loss_i = cross_entropy_loss(logits, labels, axis=0)
loss_t = cross_entropy_loss(logits, labels, axis=1)
loss   = (loss_i + loss_t)/2
```

Figure 3. Numpy-like pseudocode for the core of an implementation of CLIP.

Outline

- CLIP: Joint Text and Image Embedding
- DALL-E-2: Generating Images from Text
- LLaVA: Multi-modality Conversation

DALLE.2: generating images from text



vibrant portrait painting of Salvador Dalí with a robotic half face



a shiba inu wearing a beret and black turtleneck



a close up of a handpalm with leaves growing from it



an espresso machine that makes coffee from human souls, artstation



panda mad scientist mixing sparkling chemicals, artstation



a corgi's head depicted as an explosion of a nebula



a dolphin in an astronaut suit on saturn, artstation



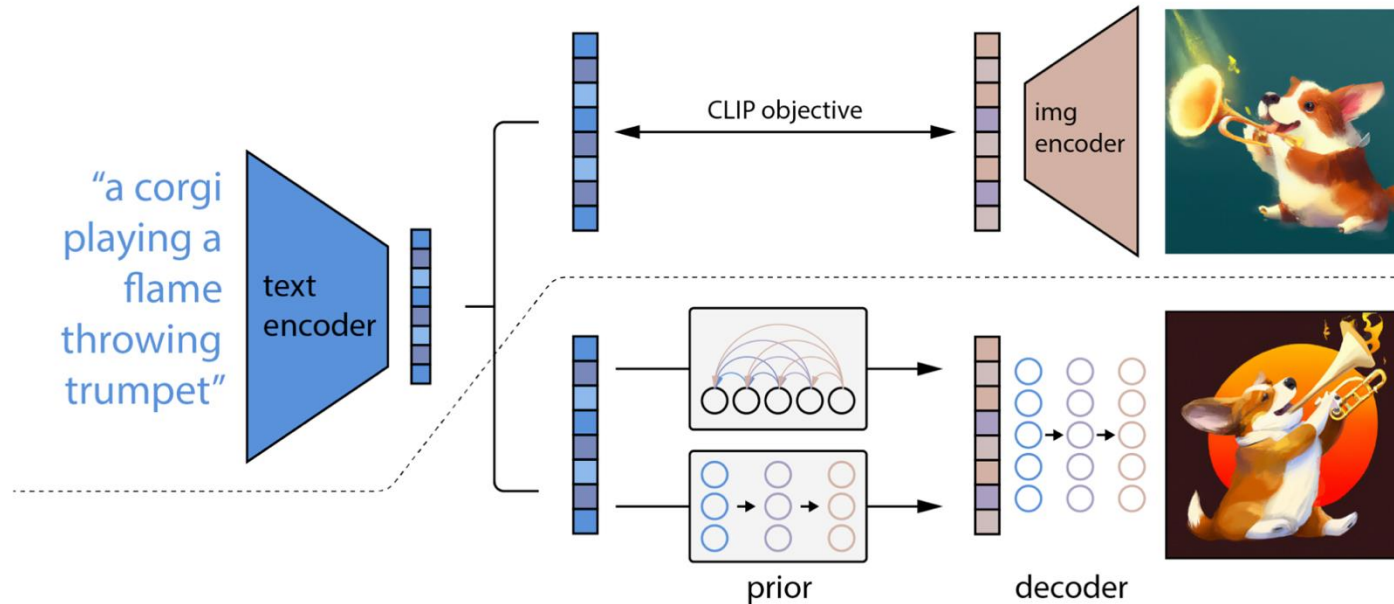
a propaganda poster depicting a cat dressed as french emperor napoleon holding a piece of cheese



a teddy bear on a skateboard in times square

Figure 1: Selected 1024×1024 samples from a production version of our model.

Model Architecture



- Learning a joint representation space for text and images based on CLIP
- A CLIP text embedding fed to an autoregressive or diffusion model to produce an image embedding
- Generate the final image with a diffusion decoder conditioning on the image embedding

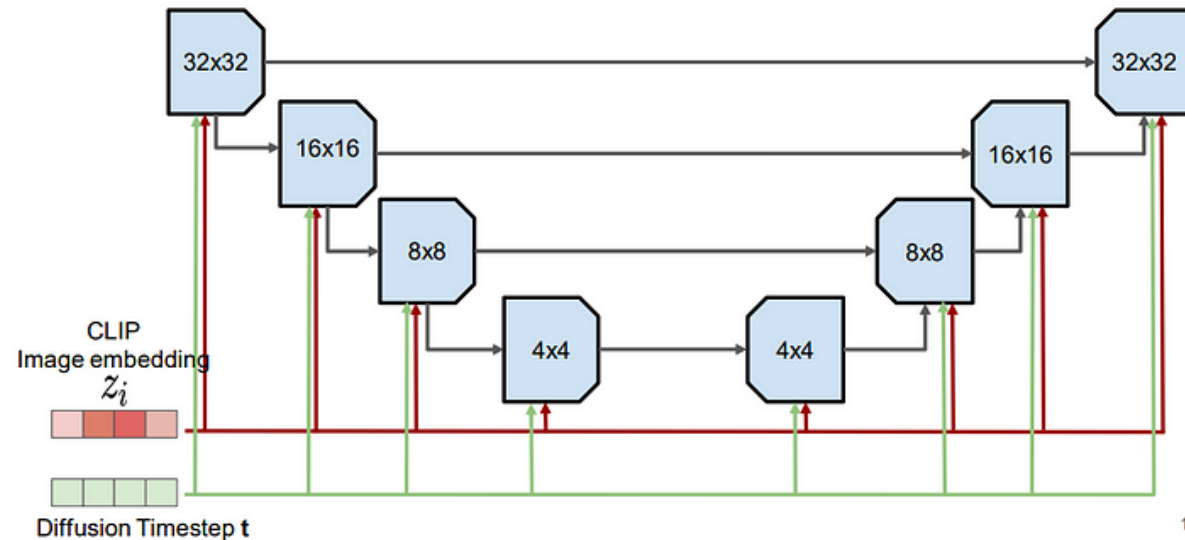
The probabilistic formulation

- Given a pair of image x and its corresponding caption y , z_i and z_t are the images and text embeddings respectively, we define the generative model as follows:
 - A *prior* $p(z_i|y)$: produces CLIP image embeddings z_i conditioned on captions y ;
 - A *decoder* $p(x|z_i, y)$: generate the final image x conditioning on z_i

$$P(x|y) = P(x, z_i|y) = P(x|z_i, y)P(z_i|y).$$

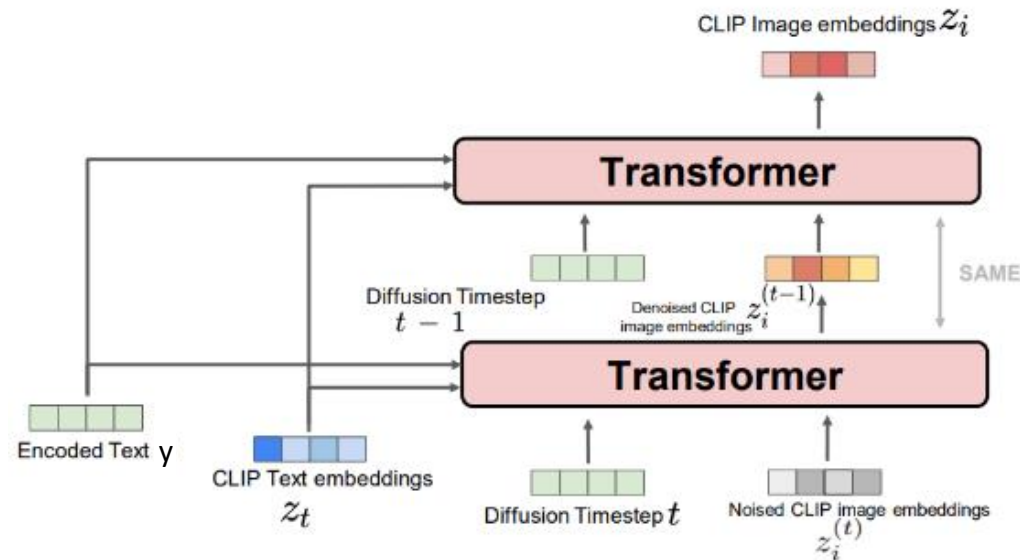
Decoder $p(x|z_i, y)$

- A U-Net diffusion model for image generation conditioning on CLIP image embedding z_i



Prior $p(z_i | y)$

- Diffusion Prior uses a diffusion model on CLIP image embedding. The input to this model is encoded text, CLIP text embedding, timestep, and noised CLIP image embedding. The continuous vector $\mathbf{z}$ is directly modelled using a Gaussian diffusion model conditioned on the caption y .



Outline

- CLIP: Joint Text and Image Embedding
- DALL-E-2: Generating Images from Text
- LLaVA: Multi-modality Conversation

LLAVA: Large Language and Vision Assistant

- Leverage GPT-4 to generate multimodal language-image instruction-following data
- Instruction tuning on such generated data and building an end-to-end trained large multimodal model:
 - Connects a vision encoder and an LLM for general purpose visual and language understanding

GPT-assisted Visual Instruction Data Generation

- Leverage ChatGPT/GPT-4 for multimodal instruction-following data collection, based on the widely existing image-text pair data.
- For each (X_v, X_c) image and caption pair, we use two types of information to prompt a text-only GPT
 - Captions X_c
 - Bounding boxes: localize the objects in the scene

Three Different Types of Instruction-Following Data

- Conversation
 - A conversation between the assistant and a person asking questions about the photo
- Detailed description
 - Include a rich and comprehensive description for an image
- Complex reasoning
 - In-depth reasoning questions

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.

Luggage surrounds a vehicle in an underground parking area

People try to fit all of their luggage in an SUV.

The sport utility vehicle is parked in the public garage, being packed for a trip

Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], backpack: [0.384, 0.696, 0.485, 0.914], suitcase: ...<omitted>

Response type 1: conversation

Question: What type of vehicle is featured in the image?

Answer: The image features a black sport utility vehicle (SUV) ...<omitted>

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip. ...<omitted>

Response type 3: complex reasoning

Question: What challenges do these people face?

Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings ...<omitted>

Table 1: One example to illustrate the instruction-following data. The top block shows the contexts such as captions and boxes used to prompt GPT, and the bottom block shows the three types of responses. Note that the visual image is not used to prompt GPT, we only show it here as a reference.

Prompts for Instruction-Following Data Generation

```
messages = [ {"role": "system", "content": f"""\nYou are an AI visual assistant, and you are seeing a single image. What you see are provided with five sentences, describing the same image you are looking at. Answer all questions as you are seeing the image.object types, counting the objects, object actions, object locations, relative positions between objects, etc. Only include questions that have definite answers:\n(1) one can see the content in the image that the question asks about and can answer confidently;\n(2) one can determine confidently from the image that it is not in the image. Do not ask any question that cannot be answered confidently.
```

Context type 1: Captions

A group of people standing outside of a black vehicle with various luggage.

Luggage surrounds a vehicle in an underground parking area

People try to fit all of their luggage in an SUV.

The sport utility vehicle is parked in the public garage, being packed for a trip

Some people with luggage near a van that is transporting it.



Context type 2: Boxes

person: [0.681, 0.242, 0.774, 0.694], person: [0.63, 0.222, 0.686, 0.516], person: [0.444, 0.233, 0.487, 0.34], backpack: [0.384, 0.696, 0.485, 0.914], backpack: [0.755, 0.413, 0.846, 0.692], suitcase: [0.758, 0.413, 0.845, 0.69], suitcase: [0.1, 0.497, 0.173, 0.579], bicycle: [0.282, 0.363, 0.327, 0.442], car: [0.786, 0.25, 0.848, 0.322], car: [0.783, 0.27, 0.827, 0.335], car: [0.86, 0.254, 0.891, 0.3], car: [0.261, 0.101, 0.787, 0.626]

Response type 1: conversation

Question: What type of vehicle is featured in the image?

Answer: The image features a black sport utility vehicle (SUV).

Question: Where is the vehicle parked?

Answer: The vehicle is parked in an underground parking area, likely in a public garage.

Question: What are the people in the image doing?

Answer: The people are trying to fit all of their luggage into the SUV, likely preparing for a trip.

Response type 2: detailed description

The image is an underground parking area with a black sport utility vehicle (SUV) parked. There are three people in the scene, with one person standing closer to the left side of the vehicle, another person in the middle, and the third person on the right side. They are all working together to pack their luggage into the SUV for a trip.

In the parking area, various luggage items are scattered around the vehicle. There are two backpacks, one located near the left rear wheel and the other closer to the right side of the vehicle. Additionally, there are two suitcases, one on the right side of the car and another further away near the center of the parking area. A bicycle can also be seen on the left side of the vehicle.

Other cars are parked around the main SUV, with one car positioned behind it and slightly to the left, another behind and slightly to the right, and the third car further behind on the right side.

Response type 3: complex reasoning

Question: What challenges do these people face?

Answer: In the image, a group of people is standing outside a black SUV in a parking area, surrounded by various pieces of luggage, including suitcases and backpacks. They are facing the challenge of fitting all their luggage into the black SUV. There are multiple suitcases and backpacks to be packed, which suggests that the group has a significant amount of belongings to accommodate. They might have to strategize and arrange the luggage efficiently to ensure that everything fits properly into the vehicle. Additionally, they need to consider the comfort of the passengers and visibility while driving, so the placement of the luggage must not obstruct the driver's view or make the passengers uncomfortable during the trip.

One example to illustrate the instruction-following data. The top block shows the contexts such as captions and boxes used to prompt GPT, and the bottom block shows the three types of responses. Note that the visual image is not used to prompt GPT, we only show it here as a reference.

Visual Instruction Tuning Architecture

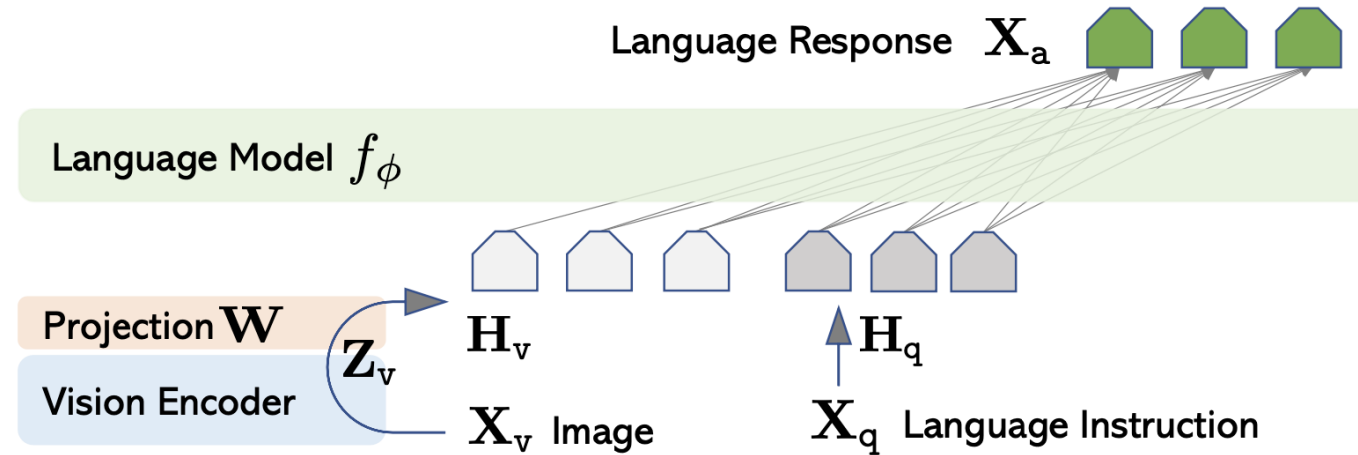


Figure 1: LLaVA network architecture.

- f_ϕ : an LLM with parameters ϕ ;
- A pretrained CLIP visual encoder $Z_v = g(X_v)$

Training

- For each image X_v , a multi-turn conversation data $(X_q^1, X_a^1, \dots, X_q^T, X_a^T)$, where T is the total number of turns. The instruction $X_{instruct}^t$ at the t -th turn is:

$$\mathbf{X}_{instruct}^t = \begin{cases} \text{Randomly choose } [\mathbf{X}_q^1, \mathbf{X}_v] \text{ or } [\mathbf{X}_v, \mathbf{X}_q^1], & \text{the first turn } t = 1 \\ \mathbf{X}_q^t, & \text{the remaining turns } t > 1 \end{cases}$$

Training

- For a sequence of length L , the probability of the target answers X_a by:

$$p(\mathbf{X}_a | \mathbf{X}_v, \mathbf{X}_{\text{instruct}}) = \prod_{i=1}^L p_{\theta}(\mathbf{x}_i | \mathbf{X}_v, \mathbf{X}_{\text{instruct}, < i}, \mathbf{X}_{a, < i}),$$

- Where θ is the trainable parameters
- $X_{\text{instruct}, < i}$ and $X_{a, < i}$ are the instruction and answer tokens in all turns before the current prediction token x_i

```
X_system-message <STOP>  
Human : X_instruct1 <STOP> Assistant: X_a1 <STOP>  
Human : X_instruct2 <STOP> Assistant: X_a2 <STOP> ...
```

The model is trained to predict the assistant answers and where to stop, and thus only **green sequence/tokens** are used to compute the loss in the auto-regressive model.

Two-stage Instruction-tuning Procedures

- Stage 1: Pre-training for Feature Alignment with single-turn conversations
 - Goal: align the CLIP imaging embedding space and LLM word embedding space
- Data: expand an image-text pair to its instruction-following version is
Human: X_q, X_v <STOP> Assistant : X_c <STOP>
- Training:
 - Keep both the visual encoder and LLM weights frozen
 - Maximize the likelihood w.r.t. the projection matrix W only. The trainable parameters are $\theta = W$

Two-stage Instruction-tuning Procedures

- Stage 2: Fine-tuning End-to-End: always keep the visual encoder weights frozen, update the pre-trained weights of the projection layer W and LLM in LLaVA ϕ . The trainable parameters are:

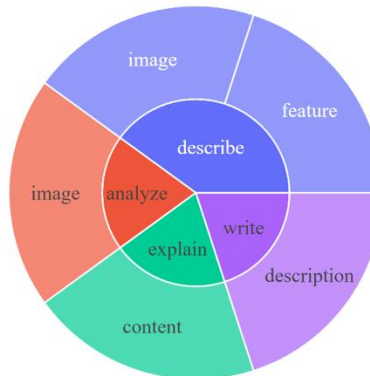
$$\theta = \{W, \phi\}$$

Application: Multimodal ChatBot

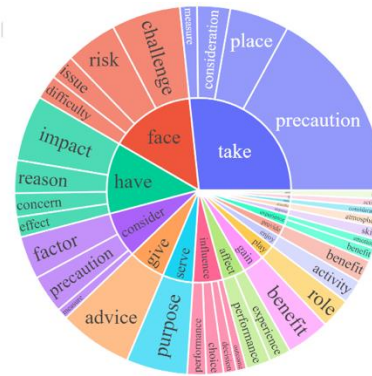
- Fine-tuning on the 158K language-image instruction-following data.



Instruction: Conversation [0, 20, 50]



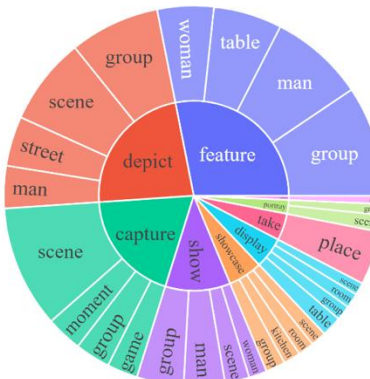
Instruction: Detailed Description [0]



Instruction: Complex Reasoning [0, 20, 50]



Response: Conversation [0, 20, 50]



Response: Detailed Description [0, 20, 50]



Response: Complex Reasoning [0, 20, 50]

Visual input example, Extreme Ironing:



Source: <https://www.barnorama.com/wp-content/uploads/2016/12/03-Confusing-Pictures.jpg>

User

What is unusual about this image?

LLaVA

The unusual aspect of this image is a man ironing clothes on the back of a minivan or van. This is not a typical place to perform this activity, as one would usually iron clothes in a more stationary and safe location, such as a home, using a regular ironing board. The scene depicted in the image is peculiar as it involves a makeshift ironing setup on a vehicle, which can be both unsafe and unconventional. Additionally, it is not clear how the man is able to maintain balance and stability while ironing clothes in such an unstable environment.

[Start a new conversation, and clear the history]

User

What's happening in the scene?

LLaVA

The image depicts a man standing on top of a yellow SUV in a busy city street. He is holding a portable ladder, seemingly ironing clothes while standing on the vehicle. Around the scene, there are other cars, a traffic light, and a fire hydrant. Additionally, there are two other people in the vicinity, one close to the man on the car and another further away. The man on the ladder appears to be attracting attention **due to his unconventional choice of ironing his clothes on top of a moving car**. The city street around him is bustling with activity, adding to the unique nature of the scene.

User

What is unusual about this image?

GPT-4 [36]

The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi.

User

What is unusual about this image?

BLIP-2

a man is sitting on the back of a yellow cab

User

What is unusual about this image?

OpenFlamingo

The man is drying his clothes on the hood of his car.

Application: Science Question Answering

- ScienceQA dataset contains 21k multimodal multiple choice questions with rich domain diversity across 3 subjects, 26 topics, 127 categories, and 379 skills.

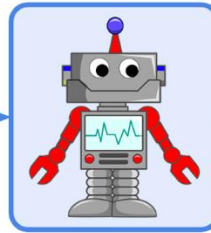
Biology Genes to traits Classification Adaptations Traits and heredity Ecosystems Classification Scientific names Heredity Ecological interactions Cells Plants Animals Plant reproduction		Physics Materials Magnets Velocity and forces Force and motion Particle motion and energy Heat and thermal energy States of matter Kinetic and potential energy Mixture		Geography State capitals Geography Maps Oceania: geography Physical Geography The Americas: geography Oceans and continents Cities States		History Colonial America English colonies in North America The American Revolution		Civics Social skills Government The Constitution	
		Chemistry Solutions Physical and chemical change Atoms and molecules Chemical reactions		Writing Strategies Supporting arguments Sentences, fragments, and run-ons Word usage and nuance Creative techniques Audience, purpose, and tone Pronouns and antecedents Persuasive strategies Editing and revising Visual elements Opinion writing		World History Greece Ancient Mesopotamia World religions American history Medieval Asia		Economics Basic economic principles Supply and demand Banking and finance	
		Engineering Designing experiments Engineering practices				Global Studies Society and environment		Verbs Verb tense	
		Units and Measurement Weather and climate				Vocabulary Categories Shades of meaning Comprehension strategies Context clues		Capitalization Formatting	Punctuation Fragments
Earth Science Weather and climate Rocks and minerals Astronomy Fossils Earth events Plate tectonics						Grammar Sentences and fragments Phrases and clauses		Phonology Rhyming	
						Figurative Language Literary devices		Reference Research skills	

<https://scienceqa.github.io/>

Question: Which type of force from the baby's hand opens the cabinet door?

Options: (A) pull (B) push

Context: A baby wants to know what is inside of a cabinet. Her hand applies a force to the door, and the door opens.



Answer: The answer is A.

BECAUSE:



Lecture: A force is a **push** or a **pull** that one object applies to a second object. The direction of a push is **away from** the object that is pushing. The direction of a **pull** is **toward** the object that is pulling.



Explanation: The **baby's hand** applies **a force** to the **cabinet door**. This force causes the **door** to **open**. The direction of this force is **toward** the **baby's hand**. This force is a **pull**.

Interacting with CLIP

- <https://colab.research.google.com/drive/1NRdzNsk58h9APO-Wo7ot8UCRX9kWxMHg?usp=sharing>
- <https://medium.com/@elvenkim1/clip-by-openai-by-first-running-the-colab-7bbd8fc8ea07>

Thanks!